

VIJAY VENKATASAMY

CIVIL & STRUCTURAL ENGINEER

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PROFILE

Civil and structural engineer with over three years of combined consultancy and international infrastructure experience. At Francis Bradshaw Consulting I take retail, commercial, industrial and residential schemes from concept to detailed design in reinforced concrete, steel, timber and masonry, including SER-certified schemes in Scotland, and I run my own site inspections, structural appraisals and defect reporting, from domestic properties to a highway bridge. Earlier, two years with VA Tech WABAG covered EPC procurement, site supervision and design checking on major water and wastewater plants. Graduate Member of the ICE, with my Initial Professional Development signed off and my CEng Professional Review report due for submission in October 2026.

PROFESSIONAL STATUS & CERTIFICATIONS

Membership: Graduate Member, Institution of Civil Engineers (GMICE, no. 93963668). Initial Professional Development record complete, signed off by my mentor and the ICE Membership Development Officer; CEng MICE Professional Review report due for submission in October 2026, review expected early 2027. IStructE graduate membership in progress.

Certifications: CSCS card holder; asbestos awareness; Diploma in Tekla Structures (2023); 15 PMI and University of California project-management certificates; Primavera P6 and MS Project (2023). Full UK driving licence.

TECHNICAL SKILLS

Design: Reinforced concrete, structural steel, timber, masonry, composite floors, steel connections (end plates, baseplates, fin plates), foundations and underpinning, temporary works and propping, remedial design, drainage and SuDS.

Codes & standards: Eurocodes (EC2, EC3, EC5, EC6) with UK National Annexes; BS 5950, BS 5400 and BS 6399 series; BS EN 1090-2 execution classes; CS 450 and CS 454 (highway structures); Building Regulations and Building Control; Scottish Building Regulations and SER; CDM 2015; IS 456.

Software: MasterSeries, Tekla Tedds, Tekla Structures, Tekla Structural Designer, STAAD.Pro, ETABS, AutoCAD, Revit, Hilti PROFIS Engineering, MATLAB, advanced MS Excel.

Practice: Site inspections and structural appraisals of existing buildings and bridges, defect reporting and remedial design, external design checking, SER submissions, planning and feasibility support, embodied-carbon assessment (PAS 2080), BOQ and bar-bending take-off, cost and programme scheduling.

PROFESSIONAL EXPERIENCE

Graduate Structural Engineer, then Civil & Structural Engineer | Francis Bradshaw Consulting, Wilmslow, UK Nov 2024 – Present
Promoted to Civil & Structural Engineer in 2026.

- Structural design, analysis and reporting from concept to detailed design in RC, steel, timber and masonry across 40+ retail, commercial, industrial and residential projects, to CDM 2015 Regulation 9 designer duties: canopies, shopfront portal frames, mezzanines, bulkheads, bressummer replacements, floor infills, crane bases, buffer walls and underpinning.
- Sole point of contact on most projects: issue drawings and calculations directly to clients, architects and contractors, and answer Building Control, Warrant Officer and SER certifier queries.
- Lead designer on the SER-certified Fort Kinnaird retail park refurbishment in Edinburgh: glass canopy, portalisation of the second bay replacing removed shopfront bracing, an 18 m hung bulkhead, M&E supports and roller-shutter posts, to BS 5950 and EC3. Staged the SER certification so external works could start while internal design continued, saving about two weeks.
- General Inspection to CS 450 of a 13.7 m single-span composite access bridge for a local council: re-derived the original HA loading to BS 5400-2 and BD 37/01, traced an undocumented 90 kN load to the tractor axles in use, and gave a bounded 40 tonne opinion subject to remedials and a CS 454 assessment.
- Carry out solo site inspections to my own RAMS, including a trading store with the public present and presumed asbestos flagged, and a Victorian roof framed in reused ship timbers with mortice joints, in a conservation area, for which I specified the remedial works.
- Six to ten year structural appraisal of a football stadium under the Guide to Safety at Sports Grounds: prepared the asset register from the record drawings and set out the intrusive investigation needed before certification, including cores to BS EN 12504-1, in-situ strength to BS EN 13791 and proof load tests on the barrier rails.
- Design temporary works for residential and commercial alterations: propping to masonry over openings, with prop centres set from the wall self-weight against the safe working load of each head, and support to retained roofs; on a spreading roof, held the props until the new ties were tensioned.
- On a house extension beside a retained oak, set foundation depths to NHBC 4.2 with stepped trench fill, Claymaster heave protection and mass concrete underpinning of the existing footings.
- Sized and detailed a fire-rated buffer wall about 17 m high, built into an existing steel frame in 6 m bays along the existing racking, with deflection heads.
- Designed a steel portal-frame maintenance building and reinforced concrete inspection pit for a local-authority waste depot, and authored the structural, drainage, pavement and earthworks specifications.

- Assess and alter existing structures: on a brick retail unit with concealed steelwork, kept the heavily loaded piers and replaced one intermediate pier with a slender steel column to extend the sales floor; advised a client against removing a load-bearing spine wall before propping, and set out the demolition sequence under CDM Regulation 9.
- Designed an overhead crane hoist over an existing bakery mixer platform from a combined 3D model, and RC crane-base foundations with a strap beam for an eccentric 6.5 t crane column at a glass works.
- Act as checker on external calculation packages and raise issues beyond the brief: compared two competing portal-frame schemes on an independent model, querying a baseplate clash, a bolt-count discrepancy, torsion and design life; on a mezzanine check, caught three errors corrected before construction.
- Rebuilt a twice-refused drainage condition submission: found the specified 1,900 litre tank holds only 950 litres of attenuation, confirmed it in writing with the manufacturer, and rechecked every storm duration at 1 in 100 year plus 40% climate change, discharge held below the greenfield rate.
- Produced the programme, cost range and risk assessment for a feasibility study comparing a 3,880 m² ground-bearing yard slab against a 2,739 m² portal-framed extension, covering permitted development against major planning, Biodiversity Net Gain, differential settlement and diversion of live drainage.
- Specified a cheaper, stronger steel SHS sub-frame in place of a contractor's aluminium proposal on a signage package, and cleared SER and landlord-surveyor queries through to Building Warrant.
- Developed the practice's embodied-carbon estimation tool (PAS 2080) and its first Carbon Reduction Plan, an 8.82% cut on the FY2021/22 baseline, presented internally in February 2025; also built AI-assisted defect-reporting and fee-letter tools signed off by an engineer before issue.

Junior Engineer Trainee, then Engineer - Design (Civil) | VA Tech WABAG Ltd, India and Southeast Asia Aug 2021 - Aug 2023

- Project engineer for all engineering packages on an international energy company's EPC effluent treatment plant: 66 procurement items across 29 mechanical, 26 piping and 11 instrumentation packages from vendors in 13 countries, with the client interface, letters of credit, export certification and milestone payments.
- Controlled package costs: sourced an alternative vendor and negotiated a raised price back down, and formalised three contract variations through back-to-back vendor provisions. Nominated for the company's Extra Miler award.
- Supervised construction of the 178.30 MLD Pillur III water treatment plant, Coimbatore: daily safety checks, toolbox talks and signed work permits; approved pour cards for RC works against IS 456 and stopped unsafe work when observed.
- Acted as checker on two sewage treatment plants (200 MLD and 150 MLD): reviewed RC and steel calculations and working drawings for digesters, clarifiers, dosing facilities and admin buildings before issue.

Engineering Internships | AK Consultant · Shree Constructions · VA Tech WABAG, India 2018 - 2021

- About 8 months in total covering structural calculations, AutoCAD and Tekla drafting, site supervision and quality inspection on residential (G+2/G+3) and water-treatment projects.

EDUCATION

MSc Structural Engineering | University of Leeds, UK 2023 - 2024

Merit (2:1). Dissertation: Monte Carlo and subset simulation in MATLAB to propagate seismic uncertainty through to structural failure probability. Group project: RIBA Stage 1-4 structural design of a proposed HS2 station for Leeds, with CDM 2015 compliance and an embodied-carbon assessment.

BEng (Hons) Civil Engineering | Anna University (Easwari Engineering College), Chennai, India 2017 - 2021

CGPA 7.96 / 10. Dissertation: partial replacement of cement and aggregates with industrial waste in concrete.

PROFESSIONAL DEVELOPMENT

- CPD recorded against the ICE development framework across the mandatory ethics, safety and sustainability themes.
- ICE technical webinars and IStructE and manufacturer CPD on subsidence, sustainable drainage, embodied carbon and timber design.